The Value of Lucky Hires: Match Quality Innovations and Firm Dynamics

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Motivation

- Firm-level fundamentals are central to employment growth and survival

  Blackwood et al. (2021); Sterk et al. (2021); Decker et al. (2020)...

- Structural models show firm-level productivity shocks drive labor adjustment and entry/exit decisions

  Hopenhayn and Rogerson (1993), Elsby and Michaels (2013), Cooper et al. (2007)

- **Open question:** what drives these underlying changes in firm-level productivity?

- **This paper:** firms with unexpectedly favorable match quality see higher subsequent growth and survival

Why Match Quality and the Concept of Match Quality

- **Match quality:** the match-specific productivity of a work-firm pair
  - Does not transfer with the worker to a new employer
  - Distinct from firm-level technical efficiency

- **Examples:** worker-workforce complementarities; labor-augmenting technology specific to each worker-firm pair

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- Match quality, is an important part of how we think about labor market dynamics, especially job creation and destruction Mortensen and Pissarides (1994); Burdett and Mortensen (1998),...
- Empirical analogue of the idiosyncratic productivity shock governing match formation, duration, and separation in the search literature

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- **Step 2:** Identify which worker and firm characteristics predict the match state
- **Step 3:** Net out the predictable component of match outcomes
  - Re-estimate HMM with richer predetermined worker characteristics
  - **Match innovation** = realized state probability $-$ predicted probability

Findings

- How firm-level exposure to different match innovations relates to subsequent employment growth, sales, and survival

- A 10 pp. shift from **Churning** (short, likely ends in separation) to **Early Growth** (medium duration, likely ends in promotion) raises:
  - employment growth by **4.9 pp.** (Compustat: **6.5 pp.**)
  - survival by **2.1 pp.** (Compustat: **2.7 pp.**)
  - sales by **14.3%** (Compustat)
  - effects persist beyond the first year

Literature

- Empirical firm dynamics: determinants of firm employment growth and survival

  Sterk et al. (2021), Blackwood et al. (2021), Decker et al. (2016), Davis et al. (2014)...

- This paper: new source of firm-specific time-varying shocks: match quality between firms and its workforce

- Measurement of match quality

  Bowlus (1995), Mühlbauer and Weber (2024), Koenraadt and Timmermans (2026), Woodcock (2015), Jackson (2013), Jinkins and Morin (2018)

- This paper: new method incorporating multiple dimensions of match quality

- HMM literature recovering persistent unobserved labor market heterogeneity

  Shibata (2019), Ahn et al. (2023), Hong et al. (2026)...

- This paper: focuses on match outcomes for each worker-firm pair
- Technical contribution: first to develop a three-step method extracting the unpredicted component of a realized outcome (Bartolucci et al. (2015))

Roadmap

- LinkedIn Employment History Data
- Step 1: Classification of Latent Match States
- Step 2: Characteristics of Worker and Firms in Each State
- Step 3: Construction of Match Innovations
- Effects on Firm Outcomes

LinkedIn Employment History Data from Revelio Labs

- Revelio aggregates and standardizes information from LinkedIn professional profiles
- Sample: Workers in US firms from January 2007 to December 2025
- Sample restrictions: workers with
  - At least 3 job spells
  - At least one year of observation after age 16
  - Exclude workers without education records or with education spells over 10 yrs
- Firm definition (*ultimate parent rcid*): an entity that may combine multiple subsidiaries, operating units, or establishments across different geographic locations under common ownership
- **22,874,918** workers, **3,896,164** unique firms, and **113,560,440** job spells

Comparison with QCEW

Data Example

user_id	position_numbe	ultimate_parent_rcid	msa	startdate	enddate	seniority	salary
46332577	1	10007	Detroit-Warren-Livonia MI MSA	2013-05-01	2013-08-01	1	21133.72
46332577	2	22193557	Chicago-Naperville-Joliet IL-IN-WI MSA	2014-07-01	2017-02-01	4	78127.84
46332577	3	1032933	Chicago-Naperville-Joliet IL-IN-WI MSA	2017-02-01	2019-03-01	1	44730.81
46332577	4	1032933	Chicago-Naperville-Joliet IL-IN-WI MSA	2019-03-01	2021-09-01	3	68953.29
46332577	5	1032933	Chicago-Naperville-Joliet IL-IN-WI MSA	2021-09-01	2022-09-01	4	81886.09
46332577	6	1032933	Chicago-Naperville-Joliet IL-IN-WI MSA	2022-09-01		5	155353.07
54151530.0	1	559948.0	San Francisco-Oakland-Fremont CA MSA	2018-09-01	2020-05-01	1	67791.8
54151530.0	2	1133445.0	San Francisco-Oakland-Fremont CA MSA	2020-08-01	2020-11-01	1	32740.39
54151530.0	3	1133445.0	San Francisco-Oakland-Fremont CA MSA	2020-11-01	2021-07-01	1	41391.56
54151530.0	4	1133445.0	San Jose-Sunnyvale-Santa Clara CA MSA	2021-10-01	2024-07-01	1	44655.14
54151530.0	5	1133445.0	Boston-Cambridge-Quincy MA-NH MSA	2024-08-01	2025-07-01	1	53389.83

Sample Data. Source: Revelio Labs

- Later merged with individual-level education records and demographic info

Classification of Transition Types

1. *Downgrade:* same firm, lower seniority
2. *Lateral transfer:* same firm, same seniority
3. *Upgrade:* same firm, higher seniority
4. *Pull separation:* new firm, higher seniority
5. *Push separation:* new firm, same or lower seniority
6. *Transition to education:* next activity is enrollment in education
7. *Transition to nonemployment:* no subsequent position or enrollment observed
8. *Relocation:* separation with a move to a different geographic area
9. *Ongoing:* position has not ended as of the last data update

Step 1: Classification of Match Outcomes using an HMM

Overview

- Summarize multidimensional match outcomes into a finite set of latent match states

- Estimate a discrete state-space model: Hidden Markov Model
  - Choose the number of states based on the estimated transition dynamics
  - Interpret the latent states using the state-specific measurement equations

- Why a Hidden Markov Model?
  - Allows match states to evolve dynamically over a worker's employment history
  - Captures persistence and mobility across match states through transition matrices
  - Generates a history-dependent prediction of the worker's next match state

Step 1: Hidden Markov Model Structure

Estimation equations

- **Latent states:** $U_n \in \{1, \ldots, K\}$
- **Observed signals** $Y_n = (Y_n^1, Y_n^2)$:
  - Y^1: spell duration in months: ≤ 6, $(6, 15]$, $(15, 36]$, > 36
  - Y^2: transition type: promotion, downgrade, separation to nonemployment, ...

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- **Parameters**, estimated following Bartolucci et al. (2017); Shibata (2019):
  - Emission probabilities: $\phi_{y|k} = \Pr(Y_n = y \mid U_n = k)$
  - Transition matrix: $P_{jk} = \Pr(U_n = k \mid U_{n-1} = j)$, time-invariant
  - Initial distribution: $\pi_k = \Pr(U_1 = k)$

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- **Covariates**:
  - π_k varies by year of first observation (entry-job cohort)
  - P_{jk} varies by the nonemployment gap between spells: 1) none, 2) short, 3) long (overlapping education), 4) long (nonemployment)

Step 1: Number of States and Model Selection

- Choose the number of states to balance *statistical fit* with *stable, economically meaningful patterns in employment outcomes*: $K = 7$
- Guided by four criteria:
  - *Bayesian Information Criterion*

    Figure for k = 2-40 states
    - Flattens after $K = 5$
    - $K = 7$ captures $\sim$83% of total BIC improvement
  - *Out-of-sample fit*
    - Improves with K
    - Gains beyond $K = 7$ are modest
  - *Distinct transition dynamics*
    - At $K = 7$, each state's transition profile is clearly distinct; the matrix stays full rank and well conditioned
  - *Parameter stability and economic interpretability*
    - $K = 7$ is stable across samples with distinct, interpretable states

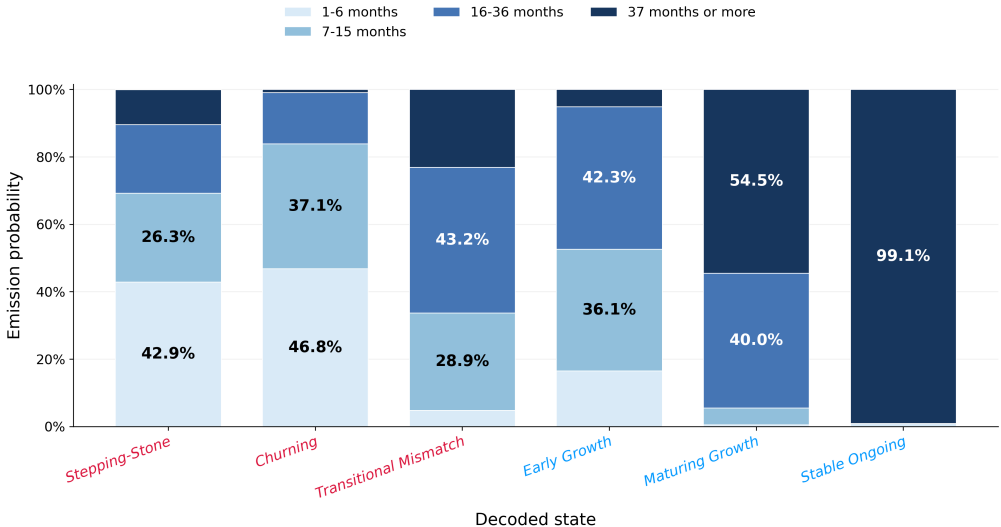
Summary of Latent Match States

- **Separation-prone states**: mainly ending in separation
  - *Stepping-Stone Job*
  - *Churning Job*
  - *Transitional Mismatch*

- **Growth-oriented states**: mainly ending in promotions/lateral moves
  - *Early Growth Track*
  - *Maturing Growth Match*
  - *Stable-Ongoing Match* (right censored)

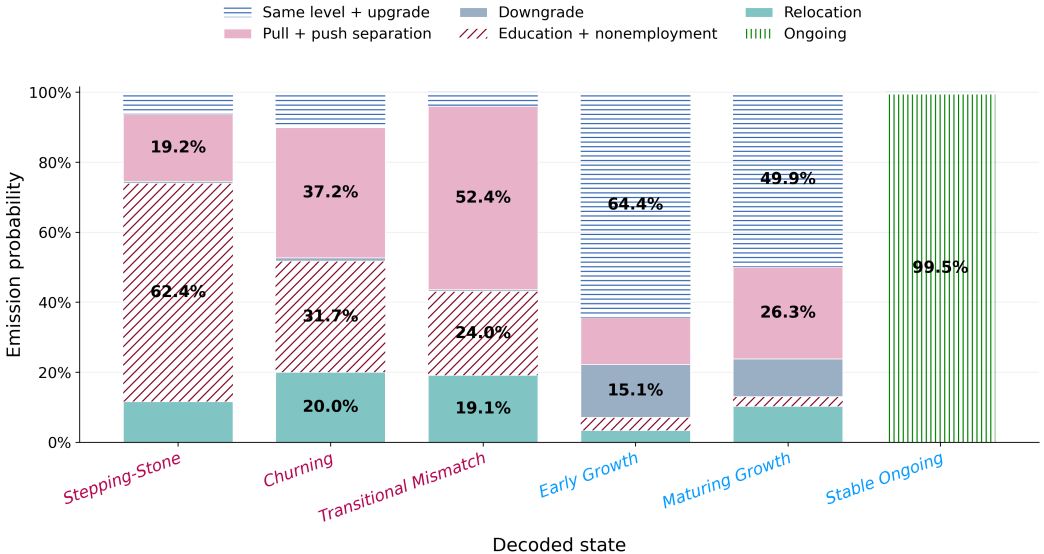
- **Censored states**
  - *Left-censored:* concentrated among first-observed spells of early cohorts

Employment Duration Probabilities

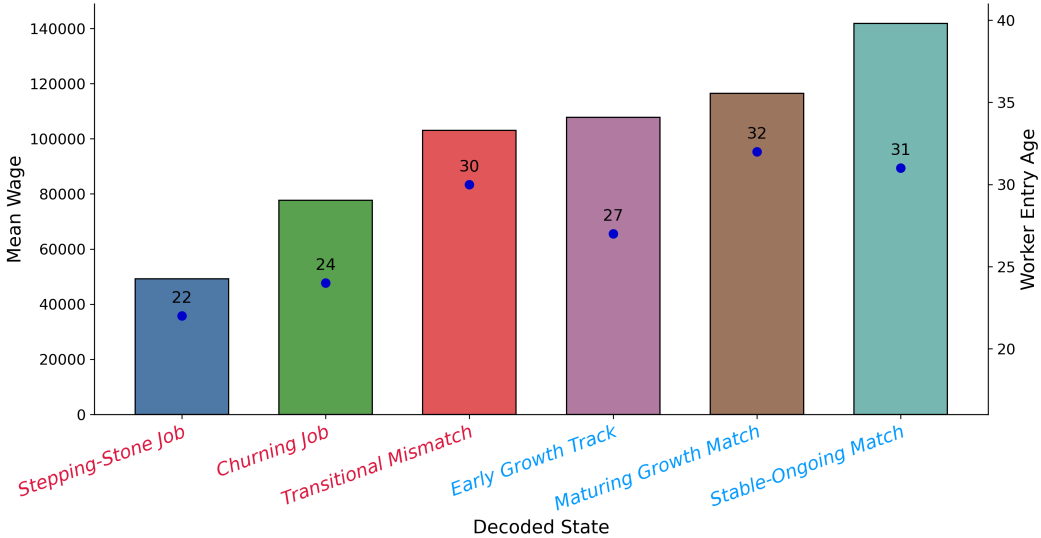


Reported probabilities; small deviations from 100% reflect rounding.

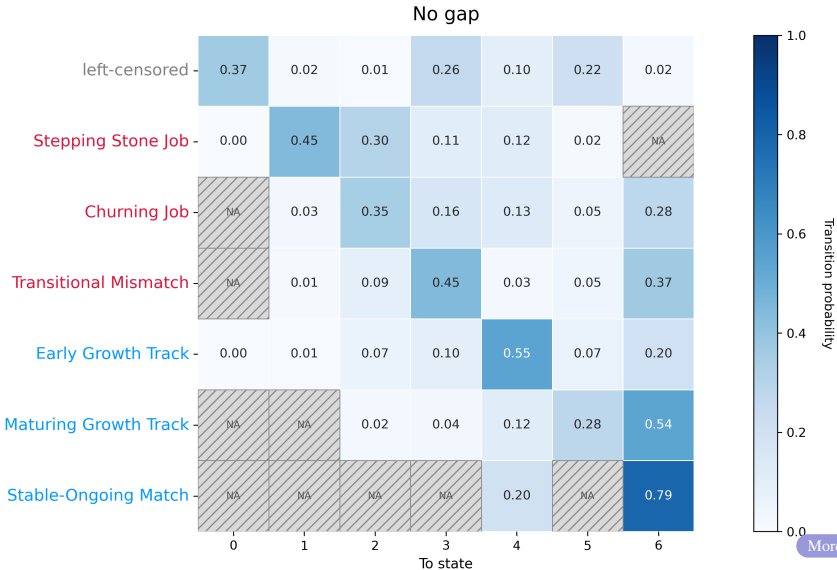
Transition Types Probabilities



Entry Wage and Age by Match State

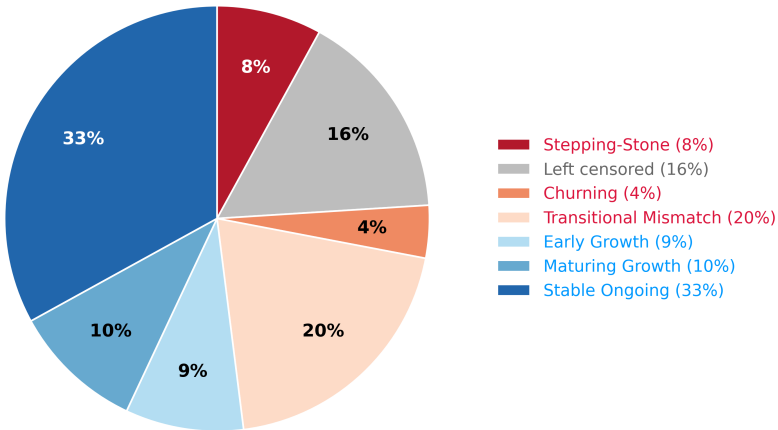


Spell-to-spell Transitions



More Transition Dynamics

Frequency of Latent States



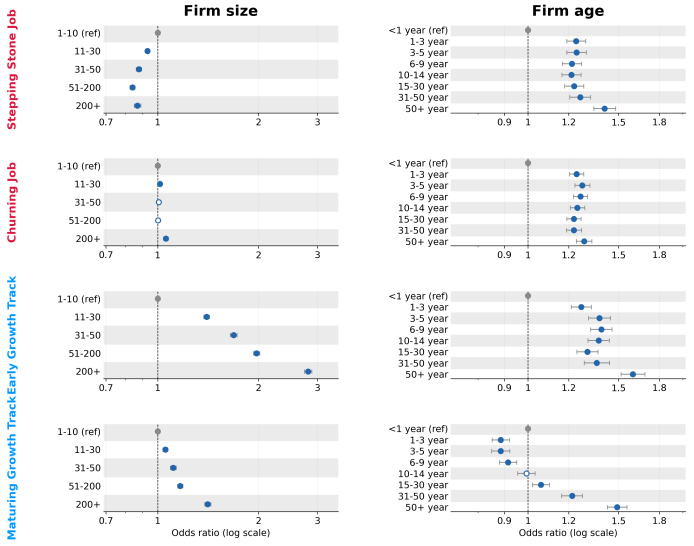
Step 2: Characteristics of Latent Match States

- Descriptive evidence from spell-level multinomial logit:
  - Who sorts into which match state, and at which firms?

Details

- **Worker characteristics**
  - Age
  - Education — Education/Age
  - Gender
  - Ethnicity — Gender/Ethnicity
- **Firm characteristics**
  - Size and age: — Size/Age
  - Sector: — Sectors

Firm Observables and Latent Match States



Main

Step 3: Construction of Spell-level Match Innovations

- **Filtered decoding:** worker i's posterior probability of state k using only information observed through position n

Validation

(1) $\pi_{i,n,k}^{\text{Filtered}} \equiv P\left(U_{i,n} = k \mid v_i, \mathbf{z}_{i,1:n}, \mathbf{Y}_{i,1:n}\right)$

v_i: entry-job cohort, $\mathbf{z}_i$: nonemployment gaps between spells, $\mathbf{Y}_i$

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v_i: entry-job cohort, $\mathbf{z}_i$: nonemployment gaps between spells, $\mathbf{Y}_i$

- **Re-estimate** transition equations $P(U_{i,n} = k \mid U_{i,n-1})$ and the initial-state $\pi_{i,k}$ with a richer set of predetermined worker characteristics:
  - age, education, gender, ethnicity, first-job cohort, nonemployment gap b/w spells
- **Match innovation** $\equiv$ realized $-$ predicted:

$$\varepsilon_{i,n,k} = \pi_{i,n,k}^{\text{Filtered}} - \hat{p}_{i,n,k}$$

  - The unpredictable component of the match outcome, given the worker's pre-match observables and employment history

Firm Exposure to Match Innovation

- For firm j in year t, exposure to match innovation for state k is

$$(2) \quad \bar{\varepsilon}_{j,t,k} = \frac{1}{N_{jt}} \sum_{(i,n) \in \mathcal{S}_{jt}} \varepsilon_{i,n,k}$$

  $\mathcal{S}_{jt}$: employment spells at firm j in year t, defined over
  (1) existing workforce or (2) new hires only

- Firm-level regression:

$$O_{jt} = \sum_{k \in \{1,2,3,5,6,7\}} \delta_k \, \bar{\varepsilon}_{j,t-1,k} + C_{j,t-1} + u_{jt}$$

  O_{jt}: employment growth rate, firm survival indicator, firm sales

- Unit of analysis: firm-year

Empirical Strategy

- Assumption 1: the predicted probability distribution of match states captures all unobserved worker heterogeneity (e.g., ability, human capital accumulation)

- Assumption 2: conditional on controls, realized match innovations are orthogonal to other time-varying firm-level determinants of subsequent outcomes
  - 6-digit NAICS, firm size, firm age + year FE
  - Firm FE + year FE
  - Firm FE + aggregated predicted probability distribution + year FE

- Remaining confounding factor: transitory firm-specific shock
  - Estimate the effects of match innovations over longer horizons

Firm Hiring and Separations by Match Type

- Both **Early Growth** and **Maturing Growth** lower separations, but only Early Growth raises hiring

Each coefficient: effect of a 10 pp. shift from Transitional Mismatch to the row's state.

	Full Sample: Stock		Compustat Firms: Stock	
	Separation	Hiring	Separation	Hiring
Stepping-stone	−1.66*	−0.01	−2.79*	0.75*
Churning	3.80*	1.16*	4.54*	0.41
Early Growth	−2.06*	0.40*	−3.29*	2.77*
Maturing Growth	−2.91*	−0.56*	−4.81*	−0.80*
Stable Ongoing	−2.68*	−0.59*	−4.05*	0.33*
N (firm-years)	15,996,088	15,996,088	86,528	86,528

Note: Omitted category is Transitional Mismatch. *Source:* Revelio Labs.

Firm Hiring and Separations by Match Type

- **Churning** leads to far more firm separations compared to **Stepping-stone**:
  - 10 pp. shift from Stepping-stone to Churning leads to **+5.5 pp.** higher separations (Compustat: **+7.3 pp.**)

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Better Early Matches Raise Firm Growth, Survival, and Sales

- Effect of a **10 pp. shift** of a firm's workforce from *Churning* to *Early Growth*:
  - **All firms:** employment growth **+4.9 pp.**, survival probability **+2.1 pp.**
  - **Compustat firms:** growth **+6.5 pp.**, survival **+2.7 pp.**, sales **+14.3%**

Effect of a 10 pp. shift from Transitional Mismatch to the targeting state.

	Full Sample: Stock		Compustat Firms: Stock[1]		
	Growth	Survival	Growth	Survival	Log(Sales)
Stepping-stone	1.42*	0.60*	2.96*	0.34	2.37
Churning	−2.71*	−0.62*	−0.95	−1.29*	−6.34*
Early Growth	2.22*	0.63*	5.36*	1.39*	8.00*
Maturing Growth	1.98*	0.36*	2.92*	0.61*	−2.22
Stable Ongoing	1.71*	0.55*	4.20*	0.15	−0.90
N (firm-years)	16,024,447	16,544,169	71,935	87,206	63,549

Variance in exposure

All Stock

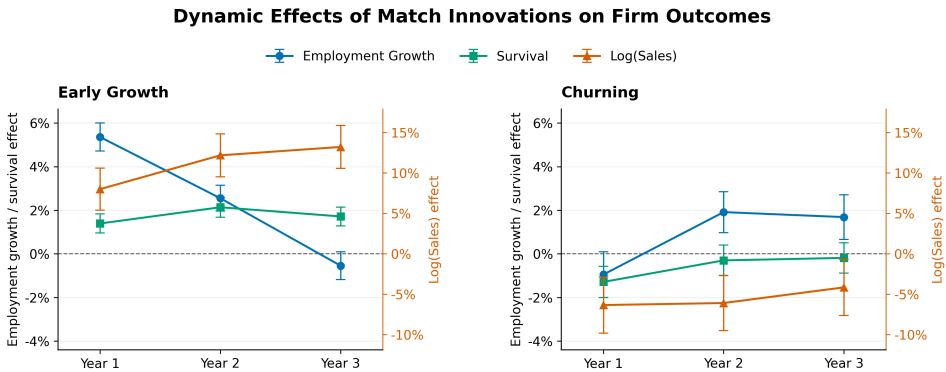
Compustat

New Hires

Note: Omitted category is Transitional Mismatch. The Churning → Early Growth effect in the bullets combines the two coefficients (0.049 = 0.022 + 0.027). *Source*: Revelio Labs, Compustat

Effects in Following Years (Compustat)

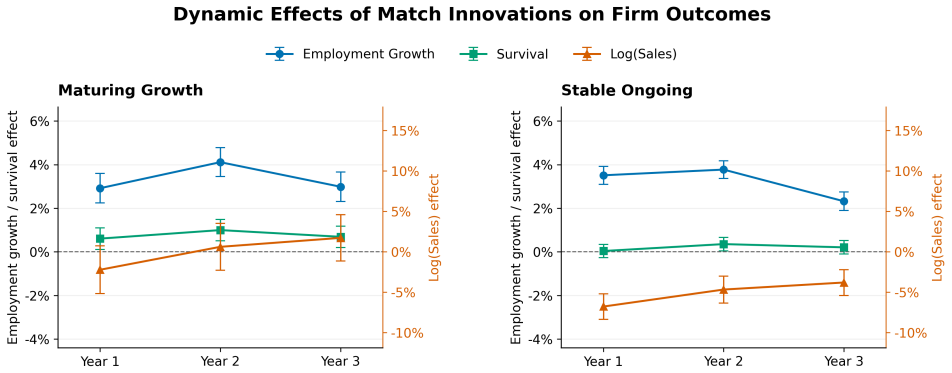
- **Early Growth:** Sales and survival effects strengthen over time
- **Churning:** negative sales effect persists through year 2; employment growth effect reverses in year 2



Error bars: approximate 90% confidence intervals. All effects are relative to Transitional Mismatch.
Growth / survival: percentage points; Log(Sales): approximate percent. Source: Revelio Labs and Compustat.

Effects in Following Years (Compustat)

- **Stable-ongoing and Maturing Growth:** employment growth effects remain strong after 3 years



Error bars: approximate 90% confidence intervals. All effects are relative to Transitional Mismatch.
Growth / survival: percentage points; Log(Sales): approximate percent. Source: Revelio Labs and Compustat.

Conclusion

- Documented that unexpected shocks to a firm's match quality carry economically meaningful information about its subsequent dynamics, beyond what is explained by predicted workforce composition and persistent firm heterogeneity.

- An unexpectedly **favorable** match leads to **better** subsequent firm outcomes, while an unexpectedly **unfavorable** match leads to **worse** subsequent firm outcomes

Comparison of Revelio Labs and QCEW by sectors

Table: Employment distribution by sectors

Source	11 Agr.	21 Min.	22 Util.	23 Constr.	31–33 Manuf.	42 Whole.	44–45 Retail
Revelio	0.2%	0.5%	0.5%	1.6%	9.9%	1.8%	6.0%
QCEW	1.0%	0.5%	0.4%	5.2%	9.7%	4.6%	12.0%
Revelio	175,949	361,782	371,004	1,232,329	7,167,188	1,329,461	4,513,772
QCEW	1,218,570	664,722	570,298	6,715,792	12,380,424	5,811,037	15,223,807
Source	48–49 Trans.	51 Info.	52 Fin.	53 Real Es.	54 Prof. Serv.	55 Mng.	56 Adm. Sup.
Revelio	1.8%	6.8%	7.2%	1.9%	9.0%	0.1%	2.5%
QCEW	4.3%	2.2%	4.6%	1.6%	6.8%	1.7%	6.7%
Revelio	1,346,893	5,115,730	5,289,857	1,495,680	6,724,540	77,523	1,941,206
QCEW	5,565,474	2,780,275	5,844,688	2,101,251	8,737,590	2,188,449	8,506,948
Source	61 Educ.	62 Health	71 Arts	72 Accom.	81 Other Serv.	92 Pub. Adm.	Unclass/Miss. Unclass/Miss.
Revelio	7.7%	6.9%	1.7%	3.0%	3.0%	7.1%	20.7%
QCEW	7.0%	13.3%	1.5%	9.7%	3.4%	3.5%	
Revelio	5,871,309	5,344,179	1,287,927	2,269,546	2,318,170	5,222,737	15,075,350
QCEW	8,919,134	17,045,222	1,964,359	12,428,279	4,343,811	4,453,579	

Emission Probabilities

	Stepping-Stone	Left censored	Churning	Transitional Mismatch	Early Growth	Maturing Growth	Stable Ongoing
Frequency	0.08	0.16	0.04	0.20	0.09	0.10	0.33
Emission 1: Employment duration ϕ^1	**1**	**2**	**3**	**4**	**5**	**6**	**7**
1–6 months	0.429	0.043	0.468	0.048	0.165	0.005	0.009
7–15 months	0.263	0.138	0.371	0.289	0.361	0.050	0.000
16–36 months	0.204	0.298	0.152	0.432	0.423	0.400	0.000
37 months or more	0.103	0.521	0.009	0.231	0.051	0.545	0.991
Emission 2: Job-transition type ϕ^2	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Education	0.367	0.105	0.094	0.000	0.034	0.017	0.000
Nonemployment	0.257	0.205	0.223	0.240	0.003	0.010	0.000
Downgrade	0.005	0.050	0.010	0.005	0.151	0.108	0.002
Same level	0.044	0.096	0.053	0.010	0.306	0.193	0.002
Upgrade	0.019	0.112	0.048	0.031	0.338	0.306	0.002
Pull separation	0.042	0.120	0.117	0.190	0.047	0.109	0.000
Push separation	0.150	0.203	0.255	0.334	0.087	0.154	0.000
Relocation	0.116	0.109	0.200	0.191	0.034	0.103	0.000
Ongoing	0.000	0.000	0.000	0.000	0.000	0.000	0.995

Step 1: Likelihood and Estimation

- The model parameters consist of four objects, $\Gamma = \{P(v), Q(z), \Phi^1, \Phi^2\}$
- Given the model parameters Γ, the likelihood function of observing individual i's employment history ($\mathbf{Y}_i$) is

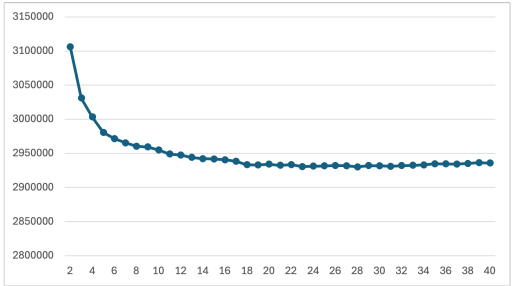
$$P(\mathbf{Y}_i \mid v_i, \mathbf{z}_i) = \sum_{\mathbf{k} \in \{1,\ldots,K\}^{N_i}} P(U_{i,1} = k_1 \mid v_i) \prod_{n=2}^{N_i} P(U_{i,n} = k_n \mid U_{i,n-1} = k_{n-1}, z_{i,n}) \times \prod_{n=1}^{N_i} \phi^1_{k_n}(Y^1_{i,n})\, \phi^2_{k_n}(Y^2_{i,n}).$$

- The likelihood of simultaneously observing labor history of all the individuals $\{Y\}_{i \in I}$ can be expressed as

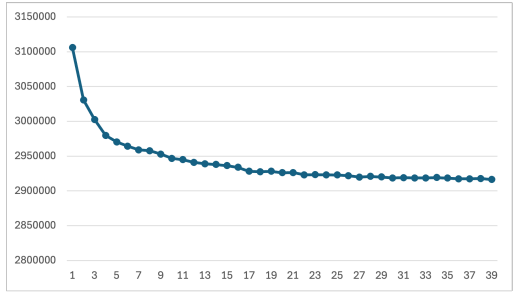
$$\sum_{i=1}^{I} \log P(\mathbf{Y}_i \mid v_i, \mathbf{z}_i; \Gamma)$$

HMM illustration

Step 1: BIC and AIC



BIC

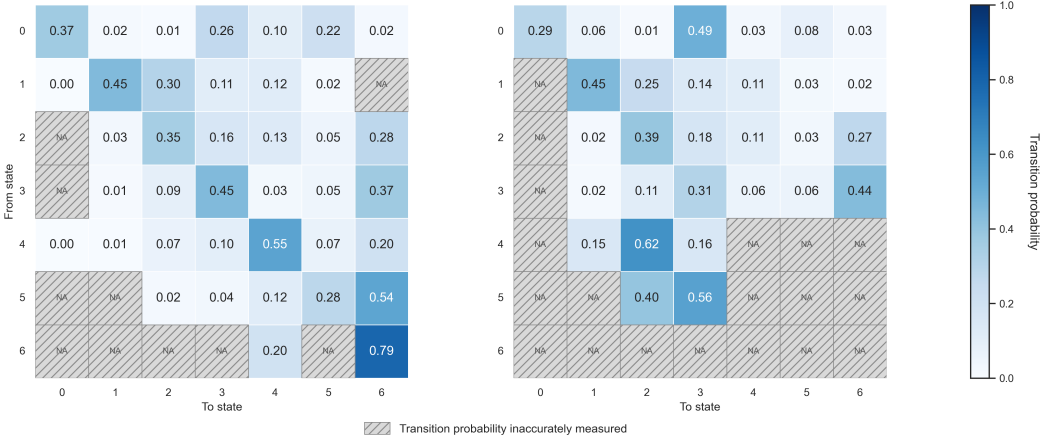


AIC

Choose the number of states

Transition Dynamics

Short Gaps

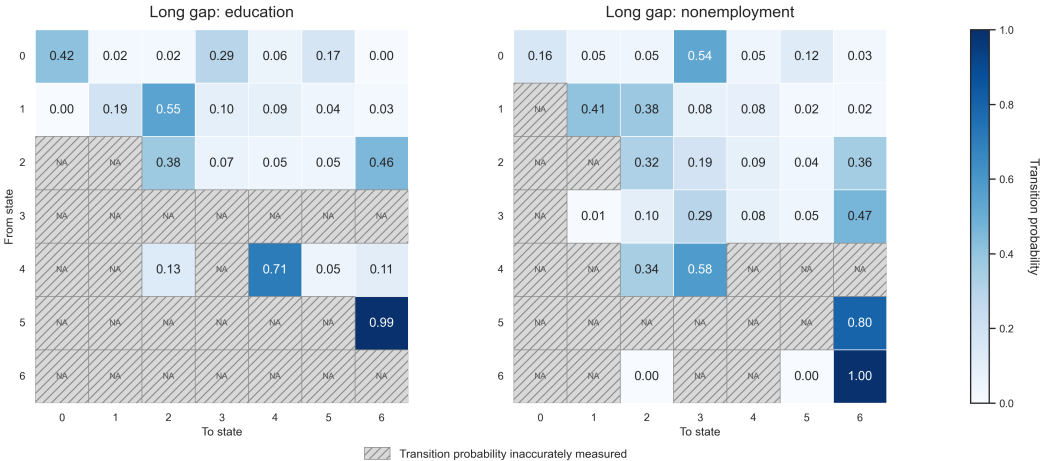


Transition Dynamics

No gap

Transition Dynamics

Long Gaps



Transition Dynamics

No gap

Step 2: Characteristics of Latent Match States

- **Local decoding:** posterior probability of latent state k for position n of worker i, conditional on the worker's full observed employment history

$$(3) \quad \pi_{i,n,k}^{Local} \equiv P\left(U_{i,n} = k \mid \mathbf{Y}_i, \mathbf{z}_i, v_i; \widehat{\Gamma}\right)$$

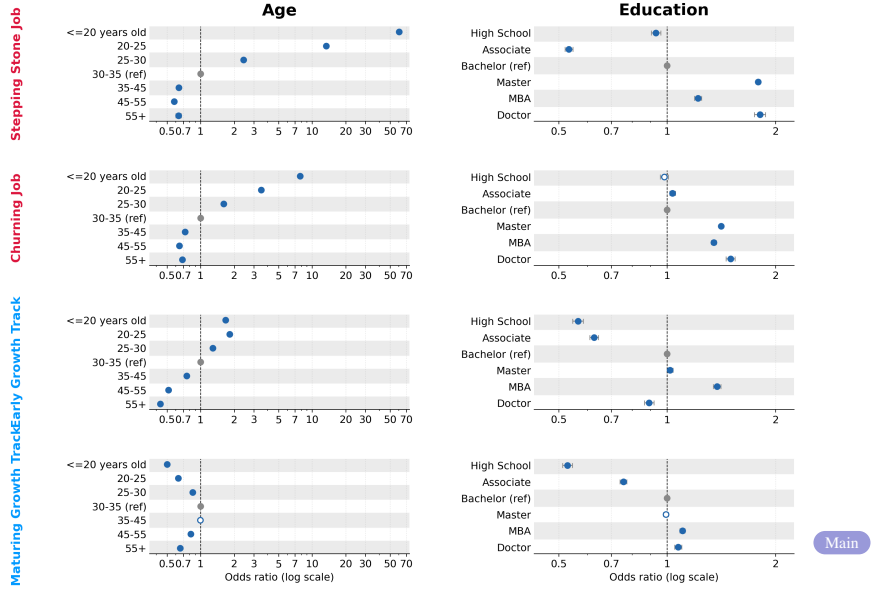
- **Spell-level multinomial logit:** how worker and firm attributes correlate with decoded match quality, relative to the baseline state (*Transitional Mismatch*, S4)

$$(4) \quad \log \frac{P(U_{ij}^t = k \mid X_{ijt})}{P(U_{ij}^t = S4 \mid X_{ijt})} = \gamma_0^k + X_{ijt}'\gamma^k, \quad k \neq S4$$

  - X_{ijt}: worker age, education, ethnicity; firm industry, size, age; year-quarter FE
  - $\exp(\gamma^k)$ = odds ratio of state k vs. baseline per covariate bin

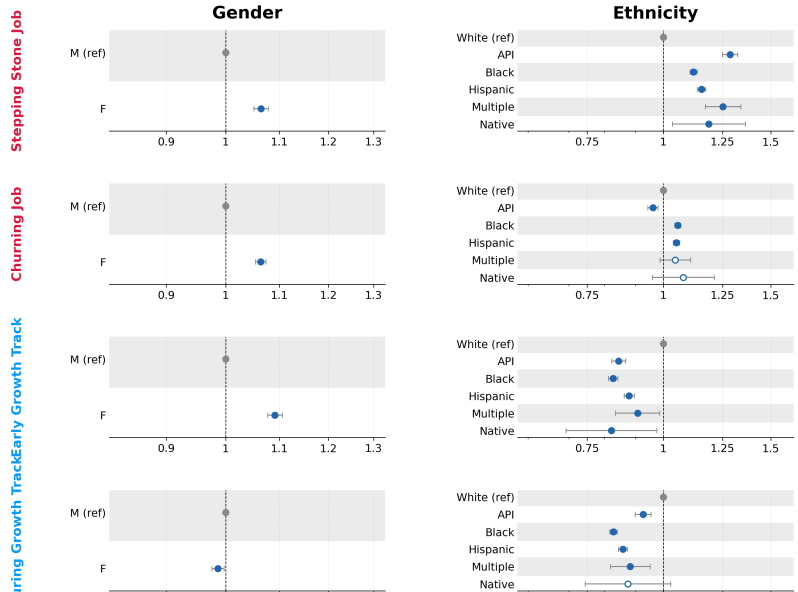
Main

Demographic Characteristics and Latent Match States



Main

Demographic Characteristics and Latent Match States



Firm Observables and Latent Match States



Main

(Non)-Persistence of Match Innovations

- All workers are equally likely to meet all firms in the market, so **potential** match quality shocks should be independent within an individual across positions
  - In search models, match quality are modeled as iid productivity shocks arriving at exogenous rates
  - But a match is formed only when the shocks is large enough

- **Realized** match quality already reflects selection by both the worker and the firm (Borovičková and Shimer, 2024)

- **Validation Question:** how much correlation remains after netting out information available prior to match formation?
  - Expect only a small fraction of persistence to remain

Match innovations

Cosine Similarity of State Probabilities and Match Innovations

Match innovations

- Random sample of 1 million workers
- *Random baseline:* replace the subsequent-spell vector with one from a different worker in the same hire-quarter, pre-employment-gap cell
- Netting out the predicted probability reduces the correlation of realized match outcomes by 55–57%
- Greater persistence remains for internal transfers, since the firm retains information about the worker across positions

	Consecutive Spells	Random Baseline	Consecutive minus Random	Number of Pairs
Match innovations				
Different firm	0.148	0.046	0.102	1,651,815
Same firm (no gap)	0.119	-0.017	0.136	815,155
State probabilities				
Different firm	0.552	0.326	0.225	1,651,815
Same firm (no gap)	0.631	0.316	0.315	815,155

Summary statistics for firm-level exposure

	Full Sample		Compustat Firms	
	Stock	New Hires	Stock	New Hires
inn_1 (Stepping-stone)	−0.041 (0.101)	0.007 (0.127)	−0.037 (0.045)	−0.004 (0.059)
inn_2 (Stable-Stagnant)	0.023 (0.102)	0.004 (0.086)	0.028 (0.055)	0.004 (0.048)
inn_3 (Churning)	−0.087 (0.112)	−0.002 (0.154)	−0.064 (0.053)	−0.004 (0.075)
inn_4 (Transitional)	−0.027 (0.215)	0.014 (0.184)	−0.004 (0.113)	0.014 (0.102)
inn_5 (Early Gr)	−0.079 (0.093)	−0.048 (0.113)	−0.048 (0.066)	0.002 (0.085)
inn_6 (Maturing Gr)	−0.009 (0.101)	−0.015 (0.082)	0.032 (0.068)	0.004 (0.058)
inn_7 (Stable Ongoing)	0.220 (0.375)	0.042 (0.295)	0.093 (0.192)	−0.016 (0.140)
N (firm-years)	17,053,441	10,083,811	87,745	81,513

- Reports the mean (standard deviation) of firm-level exposure to match innovations

- Large variance in firms' exposure across states
- Public firms show smaller variance relative to the full sample

Effect (Stock)

Match Innovations and Firm Outcomes: Stock Measure

	Growth	Survival	Growth	Survival	Growth	Survival
inn_1 (Stepping-st)	0.153***	0.115***	0.190***	0.077***	0.142***	0.060***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.001)
inn_2 (Stable/Stagnant)	0.216***	0.173***	0.298***	0.070***	0.257***	0.091***
	(0.001)	(0.001)	(0.002)	(0.001)	(0.002)	(0.001)
inn_3 (Churning)	−0.111***	−0.076***	−0.183***	−0.058***	−0.271***	−0.062***
	(0.001)	(0.001)	(0.002)	(0.001)	(0.002)	(0.001)
inn_5 (Early Gr)	0.265***	0.105***	0.224***	0.052***	0.222***	0.063***
	(0.001)	(0.000)	(0.001)	(0.000)	(0.001)	(0.000)
inn_6 (Maturing Gr)	0.245***	0.142***	0.184***	0.031***	0.198***	0.036***
	(0.001)	(0.001)	(0.002)	(0.001)	(0.002)	(0.001)
inn_7 (Stable Ongoing)	0.156***	0.143***	0.182***	0.049***	0.171***	0.055***
	(0.000)	(0.000)	(0.001)	(0.000)	(0.001)	(0.000)
Reference state	4	4	4	4	4	4
Firm size and age	✓	✓				
6-digit NAICS	✓	✓				
Firm FE			✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Predicted probabilities					✓	✓
Within R^2	0.0284	0.0822	0.0174	0.0053	0.0221	0.0109
N (firm-years)	16,482,874	17,053,441	16,024,447	16,544,169	16,024,447	16,544,169

Effect (Stock)

Match Innovations and Firm Outcomes: Compustat Firms (Stock)

	Match Innovations			Controlling for Predicted Probabilities		
	Growth	Survival	Log Sales	Growth	Survival	Log Sales
inn_1 (Stepping-st)	0.305***	0.016	0.091	0.296***	0.034	0.237
	(0.050)	(0.040)	(0.189)	(0.051)	(0.041)	(0.195)
inn_2 (Stable/Stagnant)	0.419***	−0.003	−0.347	0.420***	0.015	−0.090
	(0.049)	(0.040)	(0.213)	(0.049)	(0.040)	(0.239)
inn_3 (Churning)	−0.058	−0.117***	−0.750***	−0.095	−0.129***	−0.634***
	(0.058)	(0.042)	(0.209)	(0.064)	(0.044)	(0.211)
inn_5 (Early Gr)	0.541***	0.117***	0.622***	0.536***	0.139***	0.800***
	(0.039)	(0.026)	(0.146)	(0.039)	(0.027)	(0.159)
inn_6 (Maturing Gr)	0.289***	0.059**	−0.265	0.292***	0.061**	−0.222
	(0.041)	(0.030)	(0.181)	(0.041)	(0.030)	(0.179)
inn_7 (Stable Ongoing)	0.353***	−0.007	−0.809***	0.351***	0.004	−0.677***
	(0.024)	(0.018)	(0.095)	(0.025)	(0.018)	(0.096)
Reference state	4	4	4	4	4	4
Firm FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Predicted probabilities				✓	✓	✓
Within R^2	0.0148	0.0014	0.0122	0.0157	0.0021	0.0162
N (firm-years)	71,935	87,206	63,549	71,935	87,206	63,549

Effect (Stock)

Match Innovations and Firm Outcomes: All (New Hires)

	Match Innovations			Controlling for Predicted Probabilities		
	Growth	Survival	Log Sales	Growth	Survival	Log Sales
inn_1 (Stepping-st)	0.305***	0.016	0.091	0.296***	0.034	0.237
	(0.050)	(0.040)	(0.189)	(0.051)	(0.041)	(0.195)
inn_2 (Stable/Stagnant)	0.419***	−0.003	−0.347	0.420***	0.015	−0.090
	(0.049)	(0.040)	(0.213)	(0.049)	(0.040)	(0.239)
inn_3 (Churning)	−0.058	−0.117***	−0.750***	−0.095	−0.129***	−0.634***
	(0.058)	(0.042)	(0.209)	(0.064)	(0.044)	(0.211)
inn_5 (Early Gr)	0.541***	0.117***	0.622***	0.536***	0.139***	0.800***
	(0.039)	(0.026)	(0.146)	(0.039)	(0.027)	(0.159)
inn_6 (Maturing Gr)	0.289***	0.059**	−0.265	0.292***	0.061**	−0.222
	(0.041)	(0.030)	(0.181)	(0.041)	(0.030)	(0.179)
inn_7 (Stable Ongoing)	0.353***	−0.007	−0.809***	0.351***	0.004	−0.677***
	(0.024)	(0.018)	(0.095)	(0.025)	(0.018)	(0.096)
Reference state	4	4	4	4	4	4
Firm FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Predicted probabilities				✓	✓	✓
Within R^2	0.0148	0.0014	0.0122	0.0157	0.0021	0.0162
N (firm-years)	71,935	87,206	63,549	71,935	87,206	63,549

Effect (Stock)

One-Standard-Deviation Effects of Match Innovations on Firm Outcomes

	Full Sample: New Hires		Compustat Firms: New Hires		
	Growth	Survival	Growth	Survival	Log Sales
S1 Stepping-stone	0.014	0.004	0.009	−0.001	−0.004
S2 Stable-Stagnant	0.008	0.001	0.023	0.001	−0.024
S3 Churning	−0.023	−0.005	−0.012	−0.009	−0.032
S5 Early Growth	0.011	0.002	0.022	0.003	0.012
S6 Maturing Growth	0.015	0.002	0.010	0.001	−0.021
S7 Stable Ongoing	0.071	0.016	0.015	0.001	−0.011
N (firm-years)	8,506,274	8,671,221	67,570	80,550	59,374

Effect (Stock)

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